Tilt is the angle of the perspective on the data, while distance is the percentage of the radius of the earth (so 1.119 is 11.9% of the radius of the earth above the earth). How do you come up with such exact settings? You have two options. The first is to understand how to describe a tilted projection like this mathematically. If you have a degree in math or geography, you can look into literature for calculating this. If, like me, you don't have that kind of background, then I would suggest building a tool, using the code we explored in this chapter, to adjust the rotation, tilt, distance, and scale settings interactively. That's how I did it, and you can play with my satellite projection tool here: http://bl.ocks.org/emeeks/10173187.

Recall my advice for understanding how the Sankey layout works. Use information visualization to visualize how the functions work so that you can better understand them and find the right settings. Otherwise, you're going to need to take a course in GIS or wait for someone to write *D3.js Mapping in Action*.

Now we'll shift gears away from visualization and back to geodata structure to explore a library that was developed by Mike Bostock and is intimately tied to D3 mapping: TopoJSON.

7.4 TopoJSON data and functionality

TopoJSON (https://github.com/mbostock/topojson) is, fundamentally, three different things. First of all, it's a data standard for geographic data, and an extension of GeoJSON. Secondly, it's a library that runs in node.js to create TopoJSON-formatted files from GeoJSON files. Thirdly, it's a library that runs in JavaScript that processes TopoJSON-formatted files to create the data objects necessary to render them with libraries like D3. You won't deal with the second form at all, and you'll only examine the first in a cursory manner as you learn about rendering TopoJSON data, merging it, and using it to find a feature's neighbors.

7.4.1 TopoJSON the file format

The difference between GeoJSON files and TopoJSON files is that while GeoJSON records for each feature an array of longitude and latitude points that describe a point, line, or polygon, TopoJSON stores for each feature an array of arcs. An arc is any distinct segment of a line shared by one or more features in your dataset. The shared border between the United States and Mexico is a single arc that's referred to in the arcs array of the feature for the United States and the arcs array of the feature for Mexico.

Because most datasets have shared segments, TopoJSON often produces significantly smaller datasets. This is part of its appeal. Another part is that if you know what segments are shared, then you can do interesting things with the data, like easily calculating the neighboring features or the shared border, or merging features.

TopoJSON stores the arcs as a reference to a particular arc in a master list of arcs that defines the coordinates of that arc. You need the Topojson.js library included in any website you're using to create maps with TopoJSON, because it changes TopoJSON into a format that D3 can read and create graphics from.